Design FMEA (qualitative, simulation-signal based) - VBF-T8R2-DFMEA-01

mode	Simulation signal	S	O	Mitigation
plating (fast charge)	anode min 0.0215 V > 0 at 4C/45C (margin +21.5 mV)	high	low	keep t+ 0.40 / D 4.2e-10; no charge beyond 4C/45C without r
al rise (fast charge)	T_max 351.93 K (+33.78 K)	medium	medium	active cooling during fast charge
lyte oxidative decomposition	4.2 V upper cutoff; HOMO unchecked (real_compute=false)	medium	low	DFT endorsement deferred (Stage-5 skipped by config)
ient capacity	1C 5.0351 Ah >= nominal 5.000 Ah	high	low	area-scaled design; process control
pletion at high rate	R2 c_e diagnosis; R3 retention 0.9631	medium	low	keep transport trio

Conclusion: highest risk = plating margin (+21.5 mV, S high / O low); mitigation implemented via V_G selection. Complete FMEA N/A (beyond pure simulation boundary).